
QuiverCombinatoricsTools

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QuiverCombinatoricsTools is a SageMath package that adds combinatorial functions to *QuiverTools* to calculate symplectic leaves of quiver varieties, available here <https://github.com/emanuel-roth/QuiverCombinatoricsTools>. It adds on to the *QuiverTools* package written by Pieter Belmans, Hans Franzen and Gianni Petrella, as seen here <https://sage.quiver.tools/> and here <https://github.com/QuiverTools/QuiverTools>, so consult their documentation when needed.

To install it, make sure you have both *QuiverTools* and *QuiverCombinatoricsTools*

```
sage --pip install git+https://github.com/QuiverTools/QuiverTools.git
sage --pip install git+https://github.com/emanuel-roth/QuiverCombinatoricsTools.git
```

and then you can simply run

```
from quiver import *
from quivercombinatorics import *
```

to get started.

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GENERATING QUIVERS

Here are some functions that help generate quivers to test examples.

`quivercombinatorics.quiver_from_cartan_matrix(C)`

Returns the quiver Q given by a Cartan matrix C

INPUT:

- C – a square matrix with entries in $\mathbb{Z}$

OUTPUT: The quiver Q given by a Cartan matrix C

EXAMPLE:

```
sage: from quivercombinatorics import *
sage: C = [[2, -1, 0], [-1, 2, -2], [0, -2, 2]]
sage: quiver_from_cartan_matrix(C)
a quiver with 3 vertices and 3 arrows
```

`quivercombinatorics.random_quiver(vertices, max_arrows_per_edge)`

Returns a randomly generated quiver

INPUT:

- `vertices` – number of vertices you want in the quiver, in $\mathbb{N}_0$
- `max_arrows_per_edge` – maximum of number of arrows you want in the quiver, in $\mathbb{N}_0$

OUTPUT: A randomly generated quiver with the prescribed vertices and maximum number of edges

EXAMPLES:

```
sage: from quivercombinatorics import *
sage: random_quiver(3, 2)
a quiver with 3 vertices and 6 arrows

sage: from quivercombinatorics import *
sage: random_quiver(46, 25)
a quiver with 46 vertices and 26388 arrows
```

We notate quivers by Q , with vertices in Q_0 and edges in Q_1 .

CONSTRUCTING Σ_λ

We follow William Crawley-Boevey in [this paper](#). Given a quiver Q with n vertices, and $\lambda \in \mathbb{Z}^n$, Σ_λ is the set of $\alpha \in \mathbb{N}^n$ such that α is a positive root of Q , $\alpha \cdot \lambda = 0$, and

$$p(\alpha) > \sum_{t=1}^r p(\beta^{(t)})$$

for any decomposition $\alpha = \beta^{(1)} + \cdots + \beta^{(r)}$ with $r \geq 2$ and $\beta^{(t)}$ a positive root of Q with $\lambda \cdot \beta^{(t)} = 0$ for all t .

`Quiver.p_function(x)`

Outputs the function $p(x) = 1 - \frac{1}{2}(x, x)$, where (x, x) is the symmetrized Euler form. Note that $p(x) \geq 0$ if x is a root, and $p(x) = 0$ if and only if x is a real root

INPUT:

- x – an element of $\mathbb{Z}Q_0$

OUTPUT: $1 - \frac{1}{2}(x, x)$

EXAMPLE:

The p function evaluated at $(2, 3)$ of the doubled A2 quiver:

```
sage: from quivercombinatorics import *
sage: Q = Quiver([[0, 1], [1, 0]])
sage: Q.p_function((2, 3))
0.0
```

We write R_λ^+ to denote the set of positive roots α with $\alpha \cdot \lambda = 0$ and $\mathbb{N}R_\lambda^+$ for the set of sums of elements of R_λ^+ . Using Theorem 5.6 of Crawley-Boevey's paper.

i Theorem (Crawley-Boevey, 2001)

For $\alpha \in \mathbb{N}^n$, then $\alpha \in \Sigma_\lambda$ if and only if $0 \neq \alpha \in \mathbb{N}R_\lambda^+$ and $(\beta, \alpha - \beta) \leq -2$ whenever $\beta \in \mathbb{N}^n$ and $0 < \alpha < \beta$.

`Quiver.R_lambda_plus(l, v)`

Returns a list of elements of R_λ^+ , the set of positive roots α with $\alpha \cdot \lambda = 0$, up to the upper bound v

INPUT:

- l – an element of $\mathbb{Z}Q_0$
- v – an element of $\mathbb{N}Q_0$

OUTPUT: A list of elements of R_λ^+ , where $\mathbf{1}$ is λ , the set of positive roots α with $\alpha \cdot \lambda = 0$, up to the upper bound v

EXAMPLE:

```
sage: from quivercombinatorics import *
sage: Q = Quiver([[0, 1], [1, 0]])
sage: Q.R_lambda_plus((1, -1), (5, 5))
[(1, 1), (2, 2), (3, 3), (4, 4)]
```

We also need the helper function `N_set`.

`quivercombinatorics.N_set(S, v)`

For a list of vectors S in $\mathbb{Z}Q_0$, returns all possible sums of vectors in S as a list, up to the upper bound v . Though not directly about quivers, this is a helper function to define Σ_λ

INPUT:

- S – a list of vectors S in $\mathbb{Z}Q_0$
- v – vector in $\mathbb{Z}Q_0$

OUTPUT: A list of all possible sums of vectors in S as a list, up to the upper bound v

EXAMPLE:

```
sage: from quivercombinatorics import *
sage: N_set([[0, 1]], [0, 3])
[(0, 1), (0, 2), (0, 3)]
```

With these functions, we can define Σ_λ .

`Quiver.sigma_lambda(l, v)`

Returns a list of elements of Σ_λ , up to the upper bound v

INPUT:

- $\mathbf{l}$ – an element of $\mathbb{Z}Q_0$
- v – an element of $\mathbb{N}Q_0$

OUTPUT: A list of elements of Σ_λ , where $\mathbf{l}$ is λ , up to the upper bound v

EXAMPLE:

```
sage: from quivercombinatorics import *
sage: Q = Quiver([[0, 1], [1, 0]])
sage: Q.sigma_lambda((1, -1), (5, 5))
[(1, 1)]
```

CB-DECOMPOSITIONS

A representation type of x of a quiver Q is a sequence $\tau = (\beta^{(1)}, n_1; \dots; \beta^{(k)}, n_k)$ with $\beta^{(i)} \in \Sigma_\lambda$ and $n_i \in \mathbb{N}$ such that

$$x = \sum_{i=1}^k n_i \beta^{(i)}$$

We allow $\beta^{(i)}$ to occur multiple times if it is an imaginary root, i.e. $p(\beta^{(i)}) > 0$. The *CB-decomposition* of x is defined to be the unique representation type of x for which

$$p(\tau) := \sum_{i=1}^k p(\beta^{(i)})$$

is maximal. In order to construct CB-decompositions (called canonical decompositions in [this paper](#), but are *not* the same as `canonical_decomposition` from *QuiverTools*). We first need to find all representation types of x , up to a bound v , for which we need the following helper functions.

`quivercombinatorics.vector_decomposition(x, S)`

For a list of vectors S in $\mathbb{Z}Q_0$, returns all possible sums of vectors in S as a list that sum up to x . This is a helper function in order to define CB-decompositions

INPUT:

- x – a vector in $\mathbb{Z}Q_0$
- S – a list of vectors S in $\mathbb{Z}Q_0$

OUTPUT: All possible sums of vectors in S as a list that sum up to x

EXAMPLE:

```
sage: from quivercombinatorics import *
sage: vector_decomposition((4, 5), [(0, 1), (1, 0), (1, 1)])
[[[(0, 1), 1], [(1, 1), 4]],
 [[(0, 1), 2], [(1, 0), 1], [(1, 1), 3]],
 [[(0, 1), 3], [(1, 0), 2], [(1, 1), 2]],
 [[(0, 1), 4], [(1, 0), 3], [(1, 1), 1]],
 [[(0, 1), 5], [(1, 0), 4]]]
```

`quivercombinatorics.small_decomposition(v, n)`

For a vector v in $\mathbb{Z}Q_0$, it returns all possible partitions of the representation type $[v, n]$

INPUT:

- v – a vector in $\mathbb{Z}Q_0$
- n – a natural number in $\mathbb{N}$

OUTPUT: All possible partitions of the representation type $[v, n]$

EXAMPLE:

```
sage: from quivercombinatorics import *
sage: small_decomposition((1, 1), 3)
[[[(1, 1), 1], [(1, 1), 1], [(1, 1), 1]],
 [[(1, 1), 2], [(1, 1), 1]],
 [[(1, 1), 3]]]
```

With these functions, we can determine all representation types.

`Quiver.all_representation_types( $l, x$ )`

Returns a list of representation types of a quiver, with respect to x

INPUT:

- l – an element of $\mathbb{Z}Q_0$
- x – an element of $\mathbb{N}Q_0$

OUTPUT: A list of representation types of a quiver, with respect to x , and where l is λ . Each representation type is stored as a list whose elements are 2-tuples, the first element is in $\mathbb{N}Q_0$, and the second is in $\mathbb{N}$

EXAMPLES:

```
Q = Quiver([[0, 1], [1, 0]])
Q.all_representation_types((1, -1), (5, 5))
[[[(1, 1), 1], [(1, 1), 1], [(1, 1), 1], [(1, 1), 1], [(1, 1), 1],
 [(1, 1), 1], [(1, 1), 1], [(1, 1), 1], [(1, 1), 2],
 [(1, 1), 1], [(1, 1), 1], [(1, 1), 3],
 [(1, 1), 1], [(1, 1), 2], [(1, 1), 2],
 [(1, 1), 1], [(1, 1), 4],
 [(1, 1), 2], [(1, 1), 3],
 [(1, 1), 5]]]

sage: C = [[2, -1, 0], [-1, 2, -2], [0, -2, 2]]
sage: Q = quiver_from_cartan_matrix(C)
sage: Q.all_representation_types((0, 0, 0), (2, 4, 3))
[[[(0, 0, 1), 1],
 [(0, 1, 0), 2],
 [(0, 1, 1), 1],
 [(0, 1, 1), 1],
 [(1, 0, 0), 2],
 [(0, 0, 1), 1], [(0, 1, 0), 2], [(0, 1, 1), 2], [(1, 0, 0), 2],
 [(0, 0, 1), 2], [(0, 1, 0), 3], [(0, 1, 1), 1], [(1, 0, 0), 2],
 [(0, 0, 1), 3], [(0, 1, 0), 4], [(1, 0, 0), 2],
 [(0, 1, 0), 1],
 [(0, 1, 1), 1],
 [(0, 1, 1), 1],
 [(0, 1, 1), 1],
 [(1, 0, 0), 2],
 [(0, 1, 0), 1], [(0, 1, 1), 1], [(0, 1, 1), 2], [(1, 0, 0), 2],
 [(0, 1, 0), 1], [(0, 1, 1), 3], [(1, 0, 0), 2],
 [(2, 4, 3), 1]]]
```

Representation types of x correspond to a symplectic leaf of the quiver variety $M_0(Q, x)$, and the symplectic leaf dimension is $2p$ applied to the representation type, where p is the `p_function`.

`Quiver.symmetric_leaf_dimension( $\tau$ )`

Returns the dimension of the symplectic leaf corresponding to the representation type τ

INPUT:

- τ – a representation type, i.e., a list whose elements are 2-tuples, the first element is in $\mathbb{N}Q_0$, and the second is in $\mathbb{N}$

OUTPUT: The corresponding symplectic leaf dimension, i.e., $2p$ applied to every element of τ

EXAMPLE:

```
sage: from quivercombinatorics import *
sage: Q = Quiver([[0, 1],[1, 0]])
sage: Q.symmetric_leaf_dimension([(1, 1), 2], [(1, 1), 3])
4
```

`Quiver.CB_decomposition( $l, x$ )`

Returns the CB decomposition, which is the representation type maximizing p

INPUT:

- l – an element of $\mathbb{Z}Q_0$
- x – an element of $\mathbb{N}Q_0$

OUTPUT: The CB decomposition with respect to x , where l is λ

EXAMPLE:

```
sage: from quivercombinatorics import *
sage: Q = Quiver([[0, 1],[1, 0]])
sage: Q.CB_decomposition((1, -1), (5, 5))
[[ (1, 1), 1], [(1, 1), 1], [(1, 1), 1], [(1, 1), 1], [(1, 1), 1]]
```

Knowing the CB-decomposition, it is then easy to calculate the dimension of the quiver variety $M_0(Q, x)$, as it gives the largest (open) symplectic leaf inside $M_0(Q, x)$.

`Quiver.quiver_variety_dimension( $l, x$ )`

Returns the dimension of the quiver variety

INPUT:

- l – an element of $\mathbb{Z}Q_0$
- x – an element of $\mathbb{N}Q_0$

OUTPUT: The dimension of the corresponding quiver variety

EXAMPLE:

```
sage: from quivercombinatorics import *
sage: Q = Quiver([[0, 1],[1, 0]])
sage: Q.quiver_variety_dimension((1, -1), (5, 5))
10
```

It's often important to know the codimension 2 leaves of the quiver variety $M_0(Q, x)$, so we code this here.

`Quiver.codimension_two_leaves( $l, x$ )`

Returns the codimension 2 leaves of the quiver variety

INPUT:

- 1 – an element of $\mathbb{Z}Q_0$
- x – an element of $\mathbb{N}Q_0$

OUTPUT: The representation types of the codimension 2 leaves of the quiver variety

EXAMPLE:

```
sage: from quivercombinatorics import *
sage: Q = Quiver([[0, 1], [1, 0]])
sage: Q.codimension_two_leaves((0, 0), (5, 5))
[[[(0, 1), 1],
 [(1, 0), 1],
 [(1, 1), 1],
 [(1, 1), 1],
 [(1, 1), 1],
 [(1, 1), 1],
 [(1, 1), 1]],
 [[(1, 1), 1], [(1, 1), 1], [(1, 1), 1], [(1, 1), 2]]]
```

ext-QUIVERS

Let $\tau = (\beta^{(1)}, n_1; \dots; \beta^{(k)}, n_k)$ be a representation type of Q . The *ext-quiver* $\tilde{Q}$ associated to the symplectic leaf is the quiver with k vertices where the number of edges between vertex i and vertex j is $-(\beta^{(i)}, \beta^{(j)})$ and the number of loops at vertex i is $p(\beta^{(i)})$. The corresponding dimension vector of $\tilde{Q}$ is $\mathbf{n} = (n_1, \dots, n_k)$, and we take $\tilde{\lambda} = 0$.

`Quiver.ext_quiver` (τ)

Given a representation type, returns the ext-quiver

INPUT:

- τ – a representation type, i.e., a list whose elements are 2-tuples, the first element is in $\mathbb{N}Q_0$, and the second is in $\mathbb{N}$

OUTPUT: The ext-quiver

EXAMPLE:

```
sage: from quivercombinatorics import *
sage: Q = Quiver([[0, 1], [1, 0]])
sage: tau = [[(1, 1), 2], [(1, 1), 1], [(1, 1), 1], [(1, 1), 1]]
sage: Q.ext_quiver(tau)
a quiver with 4 vertices and 4 arrows
```

`quivercombinatorics.ext_dimension_vector` (τ)

For a representation type τ , returns the associated dimension vector

INPUT:

- τ – a representation type, i.e., a list whose elements are 2-tuples, the first element is in $\mathbb{N}Q_0$, and the second is in $\mathbb{N}$

OUTPUT: a vector in $\mathbb{N}Q_0$

EXAMPLE:

```
sage: from quivercombinatorics import *
sage: tau = [[(1, 1), 2], [(1, 1), 1], [(1, 1), 1], [(1, 1), 1]]
sage: ext_dimension_vector(tau)
(2, 1, 1, 1)
```


CLASSIFICATION OF MINIMAL DEGENERATIONS OF SYMPLECTIC LEAVES

For a quiver Q , a positive root α is *minimal imaginary* if it is imaginary and given any positive root β , $\alpha > \beta$ implies β is real.

`Quiver.is_minimal_imaginary_root(x)`

Tests whether a vector is a vector is a minimal imaginary root

INPUT:

- x – a vector in $\mathbb{Z}Q_0$

OUTPUT: `True` if x is a minimal imaginary root, and `False` otherwise

EXAMPLE:

```
sage: from quivercombinatorics import *
sage: Q = KroneckerQuiver(2)
sage: Q.is_minimal_imaginary_root((1,1))
True
```

`Quiver.all_minimal_imaginary_positive_roots(v)`

Returns all minimal imaginary positive roots up to a bound v

INPUT:

- v – a vector in $\mathbb{N}Q_0$

OUTPUT: A list of all minimal imaginary positive roots up to a bound v

EXAMPLE:

```
sage: from quivercombinatorics import *
sage: Q = KroneckerQuiver(2)
sage: Q.all_minimal_imaginary_positive_roots((3,3))
[(1,1)]
```

Given a quiver Q , a *subquiver* $T = (T_0, T_1)$ is specified by a subset $T_0 \subset Q_0$ of the vertices. Then, for $i, j \in T_0 \subset Q_0$ the number of edges between vertex i and j in T equals the number of edges between vertex i and j in Q . In particular, if $i \in T_0$ then the number of loops at i in T is the same as the number of loops at i in Q . The support of a dimension vector v is the subset $T_0 \subset Q_0$ of all vertices i such that $v_i \neq 0$. We think of the support of v as a subquiver of Q .

We say that the subquiver T is of *affine ADE type* if it is one of the graphs appearing in the classification of simply-laced affine Dynkin diagrams $\hat{A}_l$, $\hat{D}_l$ or $\hat{E}_6$, $\hat{E}_7$, $\hat{E}_8$. Then there is a unique imaginary root δ , with $p(\delta) = 1$, whose support is T . The key result is the classification of minimal imaginary roots is the following (Bellamy-Schedler 2026).

Theorem (Bellamy-Schedler, 2026)

If M is a closed leaf in $M_0(Q, x)$ and $M \subset \bar{L}$ is a minimal degeneration, then $\bar{L} \cong M \times S$, where S is isomorphic to one of the following isolated singularities:

1. A Kleinian singularity $(\mathbb{C}^2/\Gamma, 0)$.
2. The type A minimal nilpotent orbit closure $(\mathcal{O}_{\min}, 0)$ in $\mathfrak{sl}_r(\mathbb{C})$.
3. $(\mathbb{C}^{2g}/\mathbb{Z}_2, 0)$.
4. $\text{Spec}(\mathbb{C}[x] \mid \deg x \geq 2)$.

We define $\tilde{\tau}(\beta, n) := (\beta, n; e_1, \alpha_1 - n\beta_1; \dots; e_r, \alpha_r - n\beta_r)$. In each of the above cases, the leaf L from this theorem corresponds to representation types:

1. $\tilde{\tau}(\delta, n)$ for δ a minimal imaginary root on an affine ADE subquiver.
2. $\tilde{\tau}((1, 1), n)$ for a two-vertex subquiver with $t \geq 3$ edges between the vertices and no loops at the vertices.
3. $\tau = (e_i, a; e_i, a; e_1, \alpha_1; \dots; e_{i-1}, \alpha_{i-1}; e_{i+1}, \alpha_{i+1}; \dots)$ where i is a vertex with $g \geq 1$ loops and $2a = \alpha_i$.
4. $\tau = (e_i, a; e_i, b; e_1, \alpha_1; \dots; e_{i-1}, \alpha_{i-1}; e_{i+1}, \alpha_{i+1}; \dots)$ where i is a vertex with $g \geq 1$ loops and $0 < a \neq b < \alpha_i$ with $a + b = \alpha_i$.

We assign the following labels to each subminimal representation type (i.e., representation types corresponding to a minimal degeneration):

1. A_l, D_l or E_6, E_7, E_8 for affine Dynkin diagram of type $\tilde{A}_l, \tilde{D}_l$ or $\tilde{E}_6, \tilde{E}_7, \tilde{E}_8$.
2. a_{r-1} .
3. c_g .
4. m_g .

`Quiver.all_subminimal_representation_types(v)`

Returns all subminimal representation types up to a bound v , and classifies them by affine Dynkin type, or a_{r-1} , c_g, m_g

INPUT:

- v – a vector in $\mathbb{N}Q_0$

OUTPUT: A list of 2-tuples, the first element is a subminimal representation type up to a bound v , the second element is its corresponding affine Dynkin type, or a_{r-1}, c_g, m_g

EXAMPLES:

```
sage: from quivercombinatorics import *
sage: Q = CyclicQuiver(3)
sage: Q.all_subminimal_representation_types((3, 3, 3))
[[[(0, 0, 1), 2], [(0, 1, 0), 2], [(1, 0, 0), 2], [(1, 1, 1), 1]], 'A_{2}'],
 [[[(0, 0, 1), 1], [(0, 1, 0), 1], [(1, 0, 0), 1], [(1, 1, 1), 2]], 'A_{2}'],
 [[[(1, 1, 1), 3]], 'A_{2}']]

sage: from quivercombinatorics import *
sage: Q = LoopQuiver(3)
sage: Q.all_minimal_non_trivial_representation_types((5))
[[[(1), 1], [(1), 4]], 'm_{3}'], [[(1), 2], [(1), 3]], 'm_{3}']
```

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```

sage: from quivercombinatorics import *
sage: A = [[0, 1, 0, 3], [1, 0, 0, 0], [0, 0, 2, 0], [0, 0, 0, 0]]
sage: Q = Quiver(A)
sage: Q.all_minimal_non_trivial_representation_types((1, 1, 4, 1))
[[[(0, 0, 0, 1), 1], [(0, 0, 1, 0), 4], [(1, 1, 0, 0), 1]], 'A_1'),
 [[(0, 0, 1, 0), 4], [(0, 1, 0, 0), 1], [(1, 0, 0, 1), 1]], 'a_{2}',
 [[(0, 0, 0, 1), 1],
 [(0, 0, 1, 0), 1],
 [(0, 0, 1, 0), 3],
 [(0, 1, 0, 0), 1],
 [(1, 0, 0, 0), 1]],
 'm_{2}'),
 [[(0, 0, 0, 1), 1],
 [(0, 0, 1, 0), 2],
 [(0, 0, 1, 0), 2],
 [(0, 1, 0, 0), 1],
 [(1, 0, 0, 0), 1]],
 'c_{2}')]

```


PLOTTING SYMPLECTIC LEAVES IN HASSE DIAGRAMS

We visualize the poset of symplectic leaves (equivalently of representation types) as a Hasse diagram: The vertices in this diagram are the representation types and the minimal degenerations, represented by an edge connecting two representation types, correspond to the situation where $L_1 \leq L_2$ (equivalently, $\tau_1 \leq \tau_2$) but there does not exist L_3 such that $L_1 < L_3 < L_2$. We wish to compute the Hasse diagram, for which there are two methods.

Method 1

Traverse the Hasse diagram from the bottom, starting with the lowest leaves. When we reach a leaf L_i , we compute the ext-quiver associated to L_i . On this ext-quiver, we compute subminimal representation types of $\mathbf{n}$. For each such representation type $\tilde{\tau} = (\gamma^{(1)}, m_1; \dots; \gamma^{(r)}, m_r)$, define:

$$D(\tilde{\tau}) := (D(\gamma^{(1)}), m_1; \dots; D(\gamma^{(r)}), m_r)$$

By (Bellamy-Schedler 2026), it is known that $D(\tilde{\tau})$ is a representation type for v .

In this way, we have a rule that assigns to each subminimal representation type of $\tilde{Q}$ a representation type $D(\tilde{\tau}(\alpha))$ of v . It is known that $\tau < D(\tilde{\tau}(\alpha))$ is a minimal degeneration and they all occur in this way. Proceeding in this way allows one to build the Hasse diagram from the bottom up.

`quivercombinatorics.D_map(m, tau)`

Evaluates a map $D : \mathbb{Z}^k \rightarrow \mathbb{Z}^n$ from dimension vectors of the ext-quiver to dimension vectors of the original quiver by $D(m_1, \dots, m_k) := \sum_{i=1}^k m_i \beta^{(i)}$

INPUT:

- `m` – a vector in $\mathbb{Z} \text{ext}(Q)_0$
- `tau` – a representation type of the original quiver, i.e., a list whose elements are 2-tuples, the first element is in $\mathbb{N}Q_0$, and the second is in $\mathbb{N}$

OUTPUT: a vector in $\mathbb{Z}Q_0$

EXAMPLE:

```
sage: from quivercombinatorics import *
sage: tau = [[(1, 1), 2], [(1, 1), 1], [(1, 1), 1], [(1, 1), 1]]
sage: D_map((1, 2, 8, -3), tau)
(8, 8)
```

`quivercombinatorics.D_map_on_rep(tau, L)`

Applies `d_map` to a representation type of the ext-quiver

INPUT:

- L – a representation type of the ext-quiver, i.e., a list whose elements are 2-tuples, the first element is in $\text{Next}(Q)_0$, and the second is in $\mathbb{N}$
- τ – a representation type, i.e., a list whose elements are 2-tuples, the first element is in $\mathbb{N}Q_0$, and the second is in $\mathbb{N}$

OUTPUT: a representation type of the ext-quiver

EXAMPLE:

```
sage: from quivercombinatorics import *
sage: tau = [[(1, 1), 2], [(1, 1), 1], [(1, 1), 1], [(1, 1), 1]]
sage: D_map_on_rep([(1, 2, 8, -3), 3], tau)
[[ (8, 8), 3 ]]
```

`Quiver.minimal_degenerations(L)`

Returns all minimal degenerations of a symplectic leaf, corresponding to the representation type L

INPUT:

- L – a representation type, i.e., a list whose elements are 2-tuples, the first element is in $\mathbb{N}Q_0$, and the second is in $\mathbb{N}$

OUTPUT: A list of representation types

EXAMPLES:

```
sage: from quivercombinatorics import *
sage: A = [[0, 1, 0, 3], [1, 0, 0, 0], [0, 0, 2, 0], [0, 0, 0, 0]]
sage: Q = Quiver(A)
sage: Q.minimal_degenerations([(0, 0, 0, 1), 1], [(0, 0, 1, 0), 4], [(1, 1, 0, 0), 1], [(0, 0, 0, 1), 1])
[[[(0, 0, 1, 0), 4], [(1, 1, 0, 1), 1]], '$a_{2}(1)$'],
 [[[(0, 0, 0, 1), 1], [(0, 0, 1, 0), 1], [(0, 0, 1, 0), 3], [(1, 1, 0, 0), 1]], '$m_{2}(1)$'],
 [[[(0, 0, 0, 1), 1], [(0, 0, 1, 0), 2], [(0, 0, 1, 0), 2], [(1, 1, 0, 0), 1]], '$c_{2}(1)$']]
```

`Quiver.get_Hasse_diagram_method_1(l, v, dimensions=False)`

Applies Method 1 to obtain data for the Hasse diagram of minimal degenerations

INPUT:

- l – an element of $\mathbb{Z}Q_0$
- v – an element of $\mathbb{N}Q_0$
- `dimensions` – When set to True, the leaves will be labeled by `(symplectic_leaf_dimension, "counter")`, where `counter` enumerates symplectic leaves of equal dimensions

OUTPUT: `all_leaves`, `elements`, `relations`, `edge_labels`, the latter three are required to construct the Hasse diagram of minimal degenerations

EXAMPLES:

```
sage: from quivercombinatorics import *
sage: C = [[2, -1, 0], [-1, 2, -2], [0, -2, 2]]
sage: Q = quiver_from_cartan_matrix(C)
sage: Q.get_Hasse_diagram_method_1((0, 0, 0), (2, 4, 3))
```

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```

([[(0, 0, 1), 1],
 [(0, 1, 0), 2],
 [(0, 1, 1), 1],
 [(0, 1, 1), 1],
 [(1, 0, 0), 2]],
 [[(0, 0, 1), 1], [(0, 1, 0), 2], [(0, 1, 1), 2], [(1, 0, 0), 2]],
 [[(0, 0, 1), 2], [(0, 1, 0), 3], [(0, 1, 1), 1], [(1, 0, 0), 2]],
 [[(0, 0, 1), 3], [(0, 1, 0), 4], [(1, 0, 0), 2]],
 [(0, 1, 0), 1],
 [(0, 1, 1), 1],
 [(0, 1, 1), 1],
 [(0, 1, 1), 1],
 [(1, 0, 0), 2]],
 [(0, 1, 0), 1], [(0, 1, 1), 1], [(0, 1, 1), 2], [(1, 0, 0), 2]],
 [(0, 1, 0), 1], [(0, 1, 1), 3], [(1, 0, 0), 2]],
 [(2, 4, 3), 1]],
 [0, 1, 2, 3, 4, 5, 6, 7],
 [(0, 4),
 (1, 5),
 (1, 0),
 (2, 0),
 (2, 5),
 (3, 2),
 (3, 1),
 (3, 6),
 (4, 7),
 (5, 4),
 (6, 5)],
 [(0, 4, '$A_{1}(1)$'),
 (1, 5, '$A_{1}(1)$'),
 (1, 0, '$c_{1}(1)$'),
 (2, 0, '$A_{1}(1)$'),
 (2, 5, '$A_{1}(1)$'),
 (3, 2, '$A_{1}(1)$'),
 (3, 1, '$A_{1}(1)$'),
 (3, 6, '$A_{1}(1)$'),
 (4, 7, '$D_{4}(1)$'),
 (5, 4, '$c_{1}(1)$'),
 (6, 5, '$m_{1}(1)$')])
    
```

`Quiver.plot_Hasse_diagram_method_1(l, v, dimensions=False)`

Constructs the Hasse diagram using Method 1, using `get_Hasse_diagram_method_1`. The output can then be plotted by appending `.plot()` for example

INPUT:

- l – an element of $\mathbb{Z}Q_0$
- v – an element of $\mathbb{N}Q_0$
- `dimensions` – When set to `True`, the leaves will be labeled by `(symplectic_leaf_dimension, counter)`, where `counter` enumerates symplectic leaves of equal dimensions

OUTPUT: The Hasse diagram as a poset

EXAMPLES:

```
sage: from quivercombinatorics import *
sage: C = [[2, -1, 0], [-1, 2, -2], [0, -2, 2]]
sage: Q = quiver_from_cartan_matrix(C)
sage: Q.plot_Hasse_diagram_method_1((0, 0, 0), (2, 4, 3))
Finite poset containing 8 elements

sage: Q = LoopQuiver(3)
sage: Q.plot_Hasse_diagram_method_1((0), (4), dimensions=True)
Finite poset containing 11 elements
```

To export this Hasse diagram as a TikZ diagram, we need to install dot2tex and graphviz.

```
sage --pip install dot2tex
sage --pip install graphviz
```

and then make sure you import the following

```
from sage.all import latex
```

`Quiver.plot_Hasse_diagram_method_1_labels(l, v, dimensions=False, latex=False)`

Constructs the Hasse diagram using Method 1, using `get_Hasse_diagram_method_1`, but also outputs the labels of the minimal degenerations. The TikZ code can be seen by applying `latex()` to the output when `latex` is set to `True`.

INPUT:

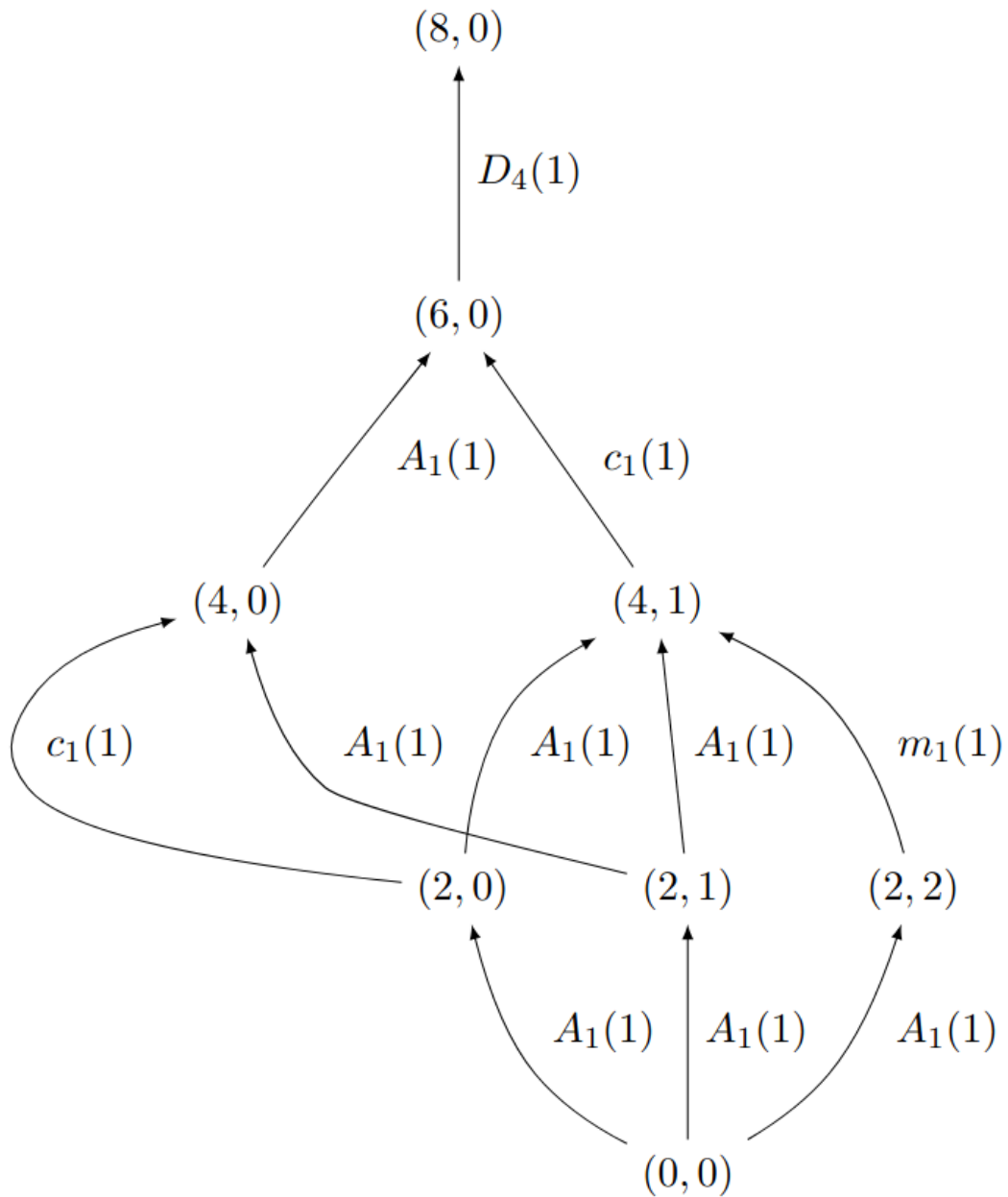
- *l* – an element of $\mathbb{Z}Q_0$
- *v* – an element of $\mathbb{N}Q_0$
- *dimensions* – When set to `True`, the leaves will be labeled by (`symplectic_leaf_dimension`, `counter`), where `counter` enumerates symplectic leaves of equal dimensions
- *latex* – When set to `True`, the labels will be formatted to be ready for a latex export

OUTPUT: The Hasse diagram as a poset, with labels

EXAMPLE:

```
sage: from quivercombinatorics import *
sage: from sage.all import latex
sage: C = [[2, -1, 0], [-1, 2, -2], [0, -2, 2]]
sage: Q = quiver_from_cartan_matrix(C)
sage: P = Q.plot_Hasse_diagram_method_1_labels((0, 0, 0), (2, 4, 3), latex=True,
→dimensions=True)
sage: P.set_latex_options(format='dot2tex', prog='dot', rankdir='up', edge_
→labels=True, color_by_label=False)
sage: tex_code = latex(P)
sage: with open("poset_diagram.tex", "w") as f:
....:     f.write(tex_code)
```

The example above should export the following diagram:



The second method to compute this Hasse diagram is as follows.

Method 2

Consider the larger set $\mathcal{D}$ of all decompositions of a given dimension vector v . Here, a decomposition is simply a way of writing v as a sum of smaller dimension vectors with multiplicities:

$$(n_1, \beta^{(1)}; \dots; n_k, \beta^{(k)}),$$

where the $\beta^{(i)}$ are dimension vectors satisfying $\beta^{(i)} \leq v$. Following [this paper](#), we say that:

$$\varepsilon = (p_1, \gamma^{(1)}; \dots; p_l, \gamma^{(l)}) > \rho = (m_1, \alpha^{(1)}; \dots; m_r, \alpha^{(r)})$$

is a *direct successor* (ε is the successor of ρ) if either:

1. $l = r + 1$ and ε can be ordered such that for all $1 \leq i \leq r - 1$ we have $(m_i, \alpha^{(i)}) = (p_i, \gamma^{(i)})$ and $m_r = p_r = p_{r+1}, \gamma^{(r)} + \gamma^{(r+1)} = \alpha^{(r)}$.
2. $l = r - 1$ and ε can be ordered such that for all $1 \leq i \leq r - 2$ we have $(m_i, \alpha^{(i)}) = (p_i, \gamma^{(i)})$ and $\alpha^{(r)} = \alpha^{(r-1)} = \gamma^{(r-1)}, m_{r-1} + m_r = p_{r-1}$.

The symplectic leaves form a subset of this set of decompositions, and by restriction, they inherit a sub-poset structure. It is shown in (Bellamy-Schedler, 2026) that this is the partial ordering given by leaf closure.

`Quiver.all_decompositions(v)`

Constructs all decompositions of a given dimension vector v

INPUT:

- v – an element of $\mathbb{N}Q_0$

OUTPUT: A list of decompositions of v

EXAMPLE:

```
sage: from quivercombinatorics import *
sage: A = [[0, 1, 0, 3], [1, 0, 0, 0], [0, 0, 2, 0], [0, 0, 0, 0]]
sage: Q = Quiver(A)
sage: Q.all_decompositions((1, 0, 2, 0))
[[[(0, 0, 1, 0), 1], [(0, 0, 1, 0), 1], [(1, 0, 0, 0), 1]],
 [[(0, 0, 1, 0), 1], [(1, 0, 1, 0), 1]],
 [[(0, 0, 1, 0), 2], [(1, 0, 0, 0), 1]],
 [[(0, 0, 2, 0), 1], [(1, 0, 0, 0), 1]],
 [[(1, 0, 2, 0), 1]]]
```

`quivercombinatorics.is_direct_successor(epsilon, rho)`

Verifies whether ϵ is the direct successor of ρ

INPUT:

- ϵ – an element of $\mathbb{N}^n$
- ρ – an element of $\mathbb{N}^n$

OUTPUT: True if ϵ is the direct successor of ρ , and False otherwise

EXAMPLE:

```
sage: from quivercombinatorics import *
sage: rho = [(1, 0), 2], [(1, 0), 3]
sage: epsilon = [(1, 0), 5]
sage: is_direct_successor(epsilon, rho)
True
```

`Quiver.get_Hasse_diagram_method_2(l, v)`

Applies Method 2 to obtain data for the Hasse diagram of minimal degenerations

INPUT:

- l – an element of $\mathbb{Z}Q_0$
- v – an element of $\mathbb{N}Q_0$
- `dimensions` – When set to `True`, the leaves will be labeled by `(symplectic_leaf_dimension, counter)`, where `counter` enumerates symplectic leaves of equal dimensions

OUTPUT: `rep_types, dimensions, rep_types_poset`, the last output is the actual poset

EXAMPLES:

```
sage: from quivercombinatorics import *
sage: C = [[2, -1, 0], [-1, 2, -2], [0, -2, 2]]
sage: Q = quiver_from_cartan_matrix(C)
sage: Q.get_Hasse_diagram_method_2((0, 0, 0), (2, 4, 3))

([[(0, 0, 1), 1],
 [(0, 1, 0), 2],
 [(0, 1, 1), 1],
 [(0, 1, 1), 1],
 [(1, 0, 0), 2]],
 [[(0, 0, 1), 1], [(0, 1, 0), 2], [(0, 1, 1), 2], [(1, 0, 0), 2]],
 [[(0, 0, 1), 2], [(0, 1, 0), 3], [(0, 1, 1), 1], [(1, 0, 0), 2]],
 [[(0, 0, 1), 3], [(0, 1, 0), 4], [(1, 0, 0), 2]],
 [[(0, 1, 0), 1],
 [(0, 1, 1), 1],
 [(0, 1, 1), 1],
 [(0, 1, 1), 1],
 [(1, 0, 0), 2]],
 [[(0, 1, 0), 1], [(0, 1, 1), 1], [(0, 1, 1), 2], [(1, 0, 0), 2]],
 [[(0, 1, 0), 1], [(0, 1, 1), 3], [(1, 0, 0), 2]],
 [[(2, 4, 3), 1]]],
 [4, 2, 2, 0, 6, 4, 2, 8],
 Finite poset containing 8 elements)
```

`Quiver.plot_Hasse_diagram_method_2( $l, v$ )`

Constructs the Hasse diagram using Method 2, using `get_Hasse_diagram_method_2`. The output can then be plotted by appending `.plot()` for example

INPUT:

- l – an element of $\mathbb{Z}Q_0$
- v – an element of $\mathbb{N}Q_0$
- `dimensions` – When set to `True`, the leaves will be labeled by `(symplectic_leaf_dimension, counter)`, where `counter` enumerates symplectic leaves of equal dimensions

OUTPUT: The Hasse diagram as a poset

EXAMPLES:

```
sage: from quivercombinatorics import *
sage: C = [[2, -1, 0], [-1, 2, -2], [0, -2, 2]]
sage: Q = quiver_from_cartan_matrix(C)
sage: Q.plot_Hasse_diagram_method_2((0, 0, 0), (2, 4, 3))
Finite poset containing 8 elements
```

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```
sage: Q = LoopQuiver(3)
sage: Q.plot_Hasse_diagram_method_2((0), (4), dimensions=True)
Finite poset containing 11 elements
```

It seems that Method 2 is slower than Method 1, so we only implemented a TikZ exporter for Method 1.